

## **Project Report**

**Title:**

**YouTube Trending Video Analytics**

**Submitted By:**

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**Tools:**

Python | Oracle SQL | Tableau

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### **1. Introduction**

This project analyzes YouTube trending videos to understand viewer engagement, identify popular content categories, evaluate sentiment in video titles, and discover country-wise and monthly trends using Python, Oracle SQL, and Tableau.

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### **2. Objectives**

- Analyze trending videos
  - Clean and preprocess data
  - Perform sentiment analysis
  - Create visualizations
  - Write SQL queries for insights
  - Build an interactive Tableau dashboard
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### **3. Data Cleaning**

- Removed duplicate records
  - Checked missing values
  - Converted date columns
  - Created Year, Month, Day, and Trending Days columns
  - Performed sentiment analysis on video titles
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### **4. Exploratory Data Analysis**

- Top Categories
- Top Channels
- Views Distribution

- Likes Distribution
  - Likes vs Views
  - Correlation Heatmap
  - Sentiment Analysis
  - Country-wise Videos
  - Monthly Trending
  - Published Day Analysis
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## 5. SQL Analysis

Performed SQL operations including:

- Aggregate Functions
  - GROUP BY
  - ORDER BY
  - Window Functions
  - Views
  - Ranking
  - Top Videos
  - Country-wise Analysis
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## 6. Tableau Dashboard

Dashboard includes:

- KPI Cards
  - Top Channels
  - Country-wise Videos
  - Likes vs Views
  - Monthly Trends
  - Sentiment Analysis
  - Filters
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## 7. Business Insights

- Trending videos are concentrated in a few categories.

- Popular channels frequently appear in trending lists.
- Higher views are associated with more likes and comments.
- Most video titles have positive or neutral sentiment.
- Publishing patterns differ across countries and months.

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## **8. Conclusion**

This project demonstrates the complete data analysis lifecycle, including data cleaning, visualization, SQL analysis, and dashboard development. It provides actionable insights into YouTube trending content and highlights proficiency in Python, Oracle SQL, and Tableau.

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